



Planning trajectories for connected and automated vehicle platoon on curved roads: A two-dimensional cooperative approach[☆]

Shengyue Yao^{a,c,1}, Yang Zhou^{b,1}, Bernhard Friedrich^a, Soyoung Ahn^{d,*}

^a Department of Transportation and Urban Engineering, Technical University of Braunschweig, Germany

^b Zachry Department of Civil and Environmental Engineering, Texas A&M, United States

^c Shanghai AI Laboratory, China

^d Department of Civil and Environmental Engineering, University of Wisconsin-Madison, United States

ABSTRACT

This paper presents a cooperative two-dimensional trajectory planning algorithm for connected and automated vehicle (CAV) platoons. Specifically, the proposed algorithm generates two-dimensional optimal trajectories for CAVs with car-following relationships cooperatively within a complex road geometry. By extending the simplified Newell's car-following model, we propose a two-dimensional Newell's car-following model as an equilibrium car-following policy for CAVs. Based on this, a multi-objective constrained optimization is systematically formulated under a Cartesian coordinate. Due to the constraint's complexity, a new solving algorithm based on the rapid random tree (RRT) technique is proposed. To test the effectiveness of our proposed models and algorithm, numerical simulation experiments with a real-world road geometry are conducted. Results indicate that our proposed method is able to generate trajectories for CAV platoons which are close to the equilibrium condition with smooth controls, while avoiding road obstacles. We further extend the definition of a one-dimensional car-following control string stability to a two-dimensional case. By this definition, we find that the proposed method can achieve empirical two-dimensional string stability, ensuring that both lateral and longitudinal disturbances are attenuated through vehicular strings.

1. Introduction

The rapid development of connected automated vehicles (CAVs) raises a growing need to navigate a platoon of CAVs in various road scenarios. This poses major technical demands on various functional modules of CAVs, including the trajectory planning module that plans the positions and kinematic dynamics of CAVs in a platoon in a cooperative manner (Chen et al., 2017; Jia & Ngoduy, 2016; Shladover et al., 2015; Yu et al., 2018; Zhao et al., 2018; Gong and Du, 2018).

A particular challenge arises from curved roads, which impose strict geometrical constraints on trajectory planning for CAVs. The curved roads are ubiquitous in urban areas. Particularly, random obstacles (e.g., road-side parking, road construction, temporary traffic control) with irregular boundaries on curved roads render further complexity; see Fig. 1 for examples. To enable CAV platoon to navigate through curved roads in complex urban settings, the trajectory planning module must be able to handle highly irregular geometrical constraints while optimizing for efficiency, safety, and string stability.

Existing CAV trajectory planning methods are predominantly based on the Frenet coordinate, which converts nonlinear geometrical constraints into linear constraints by setting a reference line (Werling et al., 2010). Further, the vehicle kinematic constraints can be transformed into polynomials in 'along' and 'lateral-offset' dimensions (Zheng et al., 2020). These features simplified the CAV

[☆] This article belongs to the Virtual Special Issue on "MLCATS".

* Corresponding author.

E-mail address: sue.ahn@wisc.edu (S. Ahn).

¹ The first two authors have equally contributed to this paper.



Fig. 1. Typical curved roads with complex urban settings. On the demonstrated roads, road-side parking vehicles generated irregular geometrical boundaries.

trajectory planning problem to be efficiently solved by Quadratic Programming (QP) based algorithms in practice (Fan et al., 2018; Zhu and Aksun-Guvenc, 2020). However, the Frenet frame requires the reference line to be sufficiently smooth, making stable transformations between the Frenet coordinate and Cartesian coordinate difficult. This could result in the planned trajectories not adequately considering kinematic dynamics, which is amplified on roads with larger curvatures. Moreover, challenges arise while coordinating trajectories between multiple CAVs within a platoon. Specifically, the kinematic interaction between CAVs and the inter-vehicle collision-avoidance constraints are challenging to incorporate in the Frenet coordinate (Li & Zhang, 2020; Li et al., 2022).

Therefore, more advanced trajectory planning methods in the Cartesian coordinate are desired, which can handle curved roads and irregular boundaries while considering kinematic coordination among CAVs. Potential approaches include optimization-based and selection-based methods. The optimization-based approaches directly model the kinematics of CAVs, with road geometries formulated as hard constraints. CAV trajectories can then be solved under iterative frameworks, such as Model Predictive Control (MPC) (Anderson et al., 2012; Ji et al., 2016; Ntousakis et al., 2016; Wang et al., 2016). Although these methods promise an adequate consideration of CAVs' kinematic constraints in a platoon, their application on curved roads with complex urban settings is difficult. It is difficult to formulate random, non-continuous boundaries. Even with a clear representation, the nonlinear and non-convex kinematic constraints severely increase computational complexity, which diminishes the feasibility of the optimization-based methods in practice.

On the contrary, the selection-based methods normally select intermediate trajectory parts (points or links) within a feasible region (geometrically and kinematically) and bridges these parts to form complete trajectories. Based on the design philosophy, these algorithms can be divided into three categories: 1) graph search methods, such as Dijkstra algorithm or A star algorithm (Anderson et al., 2012; Dolgov et al., 2010; Kala and Warwick, 2013; Miller et al., 2008), 2) interpolating curve methods, based on polynomial curves or Bezier curves (Cong et al., 2010; Perez et al., 2013) 3) sampling-based methods, such as the probabilistic roadmap method (PRM) (Kavraki et al., 1996) and the rapid random tree (RRT) (LaValle, 1998; Paden et al., 2016).

The graph search methods, interpolating curve methods and PRM, are not effective for handling irregular boundaries. Specifically, they require assumptions on known a-priori trajectory granularity, intermediate trajectory points, or curve-fitting parameters. These assumptions are difficult to tune for wide-varying road geometries, such that it is hard to ensure kinematic constraints and physical consistency. The RRT-based approach is capable of assuring universality with the property of 'Voronoi bias', by which the algorithm tends to find feasible intermediate points first in regions that have not been explored yet (Lindemann & La Valle, 2004). Another merit of the RRT-based approach is that it can incorporate constraints for vehicle kinematic characteristics. By randomly sampling the intermediate state space (e.g., position and speed), the RRT-based approach is able to find feasible states within predefined constraints and to generate trajectories by sequentially collecting found states over a time horizon that link the initial state and the terminal state within feasible road geometry. Based on these characteristics, various RRT-based algorithms have been proposed to improve computational efficiency (Kuffner & La Valle, 2000; Karaman & Frazzoli, 2011; Karaman et al., 2011) and incorporate vehicle kinematic constraints (Perez et al., 2012; Ragaglia et al., 2015; Ragaglia et al., 2016; Yuan et al., 2020). However, the cooperative kinematic dynamics and constraints of CAVs within a platoon has not been considered by existing studies.

To this end, this paper aims to develop a trajectory planning algorithm that cooperatively plans two-dimensional kinematic dynamics of CAVs in a platoon on curved roads, considering complex urban settings. The proposed algorithm will embed the two-dimensional car-following control in trajectory planning using the RRT technique. Specifically, we will extend the constant time gap policy in one-dimensional space into a two-dimensional space by extending Newell's car-following model. Based on that, a constrained LQR control is adopted for the car-following control due to its analytical nature and readiness to integrate with RRT. Our proposed algorithm significantly extends the existing RRT-based algorithms by simultaneously considering the two-dimensional car-following process and cooperation among CAVs, to satisfy driving tasks while ensuring string stability for a CAV platoon.

This paper is organized as follows. Section 2 gives a detailed problem description and model formulation. Section 3 presents the distributed framework of planning trajectories for CAV platoons. Section 4 provides the detailed solving algorithms for the CAV platoon optimal trajectory planning problem. Section 5 describes the experimental design and results. Section 6 concludes the paper

with some concluding remarks and future work.

2. Problem description and model formulation

This section gives a detailed problem description for the two-dimensional trajectory planning of a CAV platoon with a general road scenario. The problem is described under two assumptions:

- (1) A complete-information scenario is assumed in this study. Specifically, the road geometric constraints are given and can be updated on time from various data sources (e.g., HD maps, road-side radars, etc.). In addition, the road geometric constraints and planned CAV trajectories can be transmitted among CAVs via the V2X communication without excessive latency.
- (2) Two-dimensional vehicle kinematics rather than detailed vehicle dynamics (e.g., driveline dynamics) are considered for analytical tractability.

According to these assumptions, the trajectory planning problem is described below.

For CAV i in a platoon of size N_{CAV} , $\forall i \in N_{CAV}$, its kinematic state at time step t , $s_i(t)$, in a Cartesian coordinates $y-z$ plane can be uniquely defined as $s_i(t) = [p_i(t), m_i(t)]^T$, which is composed of its position state $p_i(t) = [p_{i,z}(t), p_{i,y}(t)]^T$ and motion state $m_i(t) = [v_i(t), \theta_i(t)]^T$, where $p_{i,z}(t)$ and $p_{i,y}(t)$ represent the projected positions along z and y axis at time step t , $v_i(t)$ represents the speed magnitude, and $\theta_i(t)$ represents the angle with respect to the y -axis respectively, as shown in Fig. 2-a. CAV i can automatically determine its control input $u_i(t) = [\alpha_i(t), \omega_i(t)]^T$, where $\alpha_i(t)$ and $\omega_i(t)$ represent CAV i 's acceleration and angular speed, respectively. Then CAV i 's kinematic motion can be represented by a continuous nonlinear state space system f :

$$\frac{ds_i(t)}{dt} = \begin{bmatrix} v_i(t) \cos(\theta_i(t)) \\ v_i(t) \sin(\theta_i(t)) \\ \alpha_i(t) \\ \omega_i(t) \end{bmatrix} = f(s_i(t), u_i(t)) \quad (1)$$

According to the auxiliary representation by [Stipanovic et al., \(2004\)](#), we define an auxiliary state $\xi_i(t)$, whose elements respectively represent the position and projected speed on the z -axis and y -axis by a nonlinear function $\chi(s_i(t))$, to change the nonlinear state space system to a linear time varying system. This simplifies the system analysis while maintaining the structure of the original system. The details are given as follows:

$$\xi_i(t) = \chi(s_i(t)) = [p_{i,z}(t), p_{i,y}(t), v_i(t) \cos(\theta_i(t)), v_i(t) \sin(\theta_i(t))]^T \quad (2)$$

We further define an auxiliary control input, which represents the acceleration projected on the z -axis and y -axis respectively, as $\eta_i(t) = [a_{i,z}(t), a_{i,y}(t)]^T$. It can be easily verified that $u_i(t)$ and $\eta_i(t)$ follows the relationship as below:

$$u_i(t) = M(\xi_i(t)) \eta_i(t) \quad (3-a)$$

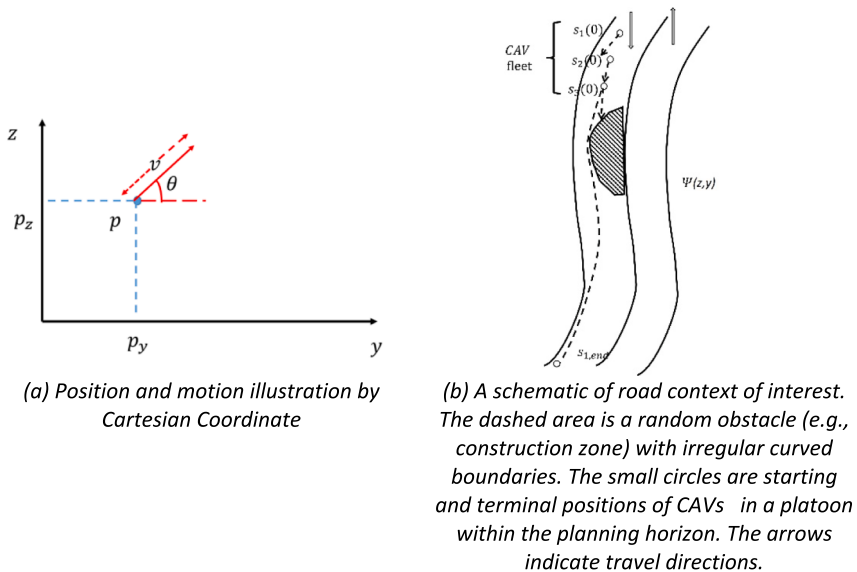


Fig. 2. Research Context Illustration.

and

$$M(\xi_i(t)) = \begin{bmatrix} \cos(\theta_i(t)) & \sin(\theta_i(t)) \\ -\sin(\theta_i(t))/v_i(t) & \cos(\theta_i(t))/v_i(t) \end{bmatrix} \quad (3-b)$$

Then the nonlinear state space system in Eq. (1) can be reformulated as a linear time-varying system by Taylor expansion, the detailed reformulation is not presented in this paper for simplicity:

$$\frac{d\xi_i(t)}{dt} = A_r \xi_i(t) + B_r \eta_i(t) \quad (4-a)$$

where

$$A_r = \begin{bmatrix} O_2 & I_2 \\ O_2 & O_2 \end{bmatrix} \quad (4-b)$$

$$B_r = \begin{bmatrix} O_2 \\ I_2 \end{bmatrix} \quad (4-c)$$

Fig. 2-B illustrates a typical scenario considered in this paper, which is a curved road with a work zone with irregular boundaries. the road context can be uniquely represented as $\Psi(z, y)$ based on the Cartesian coordinates. Given $\Psi(z, y)$ and the initial condition of CAV i , $s_i(0)$, CAV i seeks an optimal trajectory to reach its terminal condition $s_{i,end}$; i.e., a sequence of optimal states $S_i^*(T_i) = \{s_i^*(0), s_i^*(t_s), \dots, s_i^*(T_i)\}$ (or identically $\Xi_i^*(T_i) = \{\xi_i^*(0), \xi_i^*(t_s), \dots, \xi_i^*(T_i)\}$) and a sequence of optimal control inputs $U_i^*(T_i) = \{u_i^*(0), u_i^*(t_s), \dots, u_i^*(T_i)\}$ (or identically $H_i^*(T_i) = \{\eta_i^*(0), \eta_i^*(t_s), \dots, \eta_i^*(T_i)\}$) in a time horizon T_i with time interval t_s . Considering the vehicle dynamics capability, we define a control input constraint set $\mathbb{U} = [\alpha_{i,min}, \alpha_{i,max}] \times [\omega_{i,min}, \omega_{i,max}]$. Without losing generality, we set $v_i(t) \in [0, v_f]$ and $\theta_i(t) \in [-2\pi, 2\pi]$, where v_f is a pre-defined free flow speed. Hence the motion constraint set is defined as $\mathbb{M} = [0, v_f] \times [-\pi, \pi]$. In addition, CAV trajectories should lie within a physical road context $\Psi(z, y)$.

In our case, we focus on CAVs that are close enough to engage in a car-following relationship. For the first leading CAV, the trajectory planning objective is set as $J_1(\xi_1(t), \eta_1(t)) = \int_0^T 1 + \eta_1(t)^T R \eta_1(t) dt$. The objective function consists of two parts: 1 and $\eta_1(t)^T R \eta_1(t)$ to simultaneously consider the travel time and driving comfort cost. $R = \text{diag}(\gamma_z, \gamma_y)$, where γ_z, γ_y are weighting coefficients with respect to the square of acceleration on z and y axis, respectively.

As a result, the trajectory optimization problem of the first leading CAV is mathematically formulated as below:

$$(\Xi_1^*(T_1), H_1^*(T_1)) = \underset{\text{argmin}}{J_1(\xi_1(t), \eta_1(t))} \quad (5-a)$$

s.t.

$$\frac{d\xi_1(t)}{dt} = A_r \xi_1(t) + B_r \eta_1(t) \quad (5-b)$$

$$s_1(t) = \chi^{-1}(\xi_1(t)) \quad (5-c)$$

$$u_1(t) = M(s_1(t)) \eta_1(t) \quad (5-d)$$

$$Fs_1(t) \in \Psi(z, y) \quad (5-e)$$

$$Eu_1(t) \in \mathbb{U}_1 \quad (5-f)$$

$$Hs_1(t) \in \mathbb{M} \quad (5-g)$$

$$\xi_1(0) = \xi_{1,initial} \quad (5-h)$$

$$\xi_1(T_1) = \xi_{1,end} \quad (5-i)$$

Eq. (5-b) guarantees the optimization process to obey the physical law of vehicle kinematics. Eq. (5-c) and Eq. (5-d) describe the relationship among $s_1(t)$, $\xi_1(t)$, $u_1(t)$ and $\eta_1(t)$. Eq. (5-e) guarantees the trajectory point $(p_{ix}(t), p_{iy}(t))$ to lie within the road context $\Psi(z, y)$ and $F = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$. Eq. (5-d) guarantees the control input to vary within the limit. Eq. (5-g) guarantees vehicles to not violate

the speed limit, where $H = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$. Eq. (5-h) and Eq. (5-i) are the initial condition and the terminal constraint, respectively.

For CAV $i > 1$, which is in car-following mode, the trajectory for its proceeding vehicle $i-1$ needs to be considered when planning its optimal trajectory. Similar to the one-dimensional CAV car-following model (Zhou et al., 2019), we define a two-dimensional

equilibrium spacing policy by extending the one-dimensional simplified Newell's car-following model to a two-dimensional case:

$$p_i^{eq}(t) = p_{i-1}^{eq}(t - \varphi_i) - l_i(t) \quad (6)$$

where $p_i^{eq}(t)$ is the equilibrium position at time point t , and the component φ_i and $l_i(t)$ represent the response time (temporal phase shift) and the standstill spacing (spatial phase shifts) for CAV i , respectively. Note that in one-dimensional case, $l_i(t)$ is a time-invariant scalar, whereas in a two-dimensional space, it is a vector in $y-z$ plane and related to its preceding vehicle's lagged historic portfolios $p_{i-1}^{eq}(t - \varphi_i) = \{p_{i-1}^{eq}(0), p_{i-1}^{eq}(t_s), \dots, p_{i-1}^{eq}(t - \varphi_i)\}$. We first find duration $\rho_i(t)$ that CAV $i-1$ travels a scalar distance c_i , similar to one-dimensional Newell's simplified car-following model as follows:

$$\int_{t-\varphi_i-\rho_i(t)}^{t-\varphi_i} p_{i-1}^{eq}(\tau) d\tau = c_i \quad (7-a)$$

Then we have:

$$l_i(t) = \begin{bmatrix} \int_{t-\varphi_i-\rho_i(t)}^{t-\varphi_i} p_{i-1}^{eq}(\tau) \sin(\theta_i(\tau)) d\tau \\ \int_{t-\varphi_i-\rho_i(t)}^{t-\varphi_i} p_{i-1}^{eq}(\tau) \cos(\theta_i(\tau)) d\tau \end{bmatrix} \quad (7-b)$$

Hence we can have $p_i^{eq}(t) = p_{i-1}^{eq}(t - \varphi_i - \rho_i(t))$.

Fig. 3 presents an example of CAV trajectories in the proposed car-following relationship, we set $\varphi_i = 0.6s$ and $c_i = 5m$ for a better illustration. In Fig. 3, CAV 1 is the first leading CAV and CAV 2, 3, 4, 5 are the following CAVs in the platoon. According to the proposed two-dimensional car-following model, CAVs' trajectories will overlap in the $y-z$ plain, with a time latency of $\varphi_i + \rho_i$, for $i > 1$, as shown in Fig. 3.

Similarly, the equilibrium auxiliary states for CAV $i > 1$ as $\xi_i^{eq}(t) = \xi_{i-1}^{eq}(t - \varphi_i - \rho_i(t))$ and their equilibrium auxiliary control inputs as $\eta_i^{eq}(t) = \eta_{i-1}^{eq}(t - \varphi_i - \rho_i(t))$ can be derived based on Eq. (7).

Subsequently, we define a new system state $\bar{\xi}_i(t) = \xi_i(t) - \xi_i^{eq}(t)$, and new control input $\bar{\eta}_i(t) = \eta_i(t) - \eta_i^{eq}(t)$ for CAV i , $i > 1$ to formulate a state space system and describe the evolution of deviation from the equilibrium state as below:

$$\frac{d\bar{\xi}_i(t)}{dt} = A_r \bar{\xi}_i(t) + B_r \bar{\eta}_i(t) \quad (8)$$

where A_r and B_r are the same as Eq. (4-b) and (4-c).

To optimize the following CAVs' trajectories, an optimal control is systematically formulated as in Eq. (9-a) to Eq. (9-g) to regulate the spacing between CAVs, speed difference, and acceleration difference over a horizon T_i for CAV i , $i > 1$

$$(\bar{\Xi}_i^*(T_i), \bar{H}_i^*(T_i)) = \text{argmin}_{J_i}(\bar{\xi}_i(t), \bar{\eta}_i(t)) \quad (9-a)$$

s.t.

$$\frac{d\bar{\xi}_i(t)}{dt} = A_r \bar{\xi}_i(t) + B_r \bar{\eta}_i(t) \quad (9-b)$$

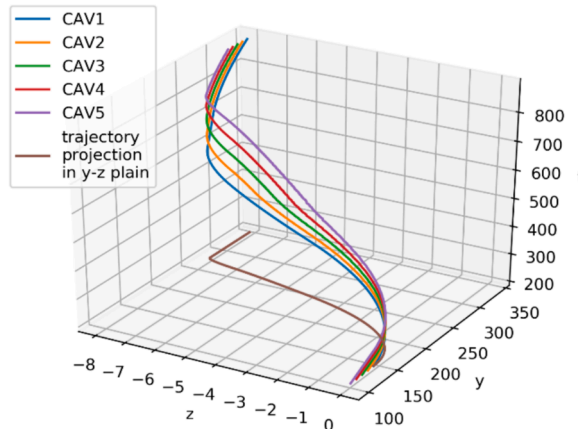


Fig. 3. CAV follower's reference trajectories based on the two-dimensional car-following model.

$$\bar{\xi}_i(t) = \xi_i(t) - \xi_i^{eq}(t) \quad (9-c)$$

$$\bar{\eta}_i(t) = \eta_i(t) - \eta_i^{eq}(t) \quad (9-d)$$

$$s_i(t) = \chi^{-1}(\xi_i(t)) \quad (9-e)$$

$$u_i(t) = M(s_i(t))\eta_i(t) \quad (9-f)$$

$$Fs_i(t) \in \Psi(z, y) \quad (9-g)$$

$$Eu_i(t) \in \mathbb{U}_i \quad (9-h)$$

$$Hs_i(t) \in \mathbb{M} \quad (9-i)$$

$$s_i(0) = s_{i,initial} \quad (9-j)$$

$$|Fs_i(t) - Fs_i(0)| - |Fs_i^{eq}(t) - Fs_i(0)| \in \Delta\mathbb{P}_i \quad (9-k)$$

In Eq. (9-a), the objective function is defined as $J_i(\bar{\xi}_i(t), \bar{\eta}_i(t)) = \bar{\xi}_i(T_i)^T Z \bar{\xi}_i(T_i) + \int_0^{T_i} \bar{\xi}_i(t)^T Q \bar{\xi}_i(t) + \bar{\eta}_i(t)^T R \bar{\eta}_i(t)$ to consider deviations from equilibrium trajectory and control input simultaneously. $Z = \text{diag}(z_1, z_2, z_3, z_4)$, $Q = \text{diag}(\beta_1, \beta_2, \beta_3, \beta_4)$, $R^f = \text{diag}(\gamma_z^f, \gamma_y^f)$ and $\beta_1, \beta_2, \beta_3, \beta_4, z_1, z_2, z_3, z_4, \gamma_z^f, \gamma_y^f$ are weighting coefficients. Eq. (9-c) and Eq. (9-d) describe deviations from the equilibrium states and control inputs, respectively. Eq. (9-e) to (9-j) has the same physical meaning as Eq. (5-b) to Eq. (5-h). Eq. (9-k) prevents the following CAV from colliding with its preceding CAV, where $|\bullet|$ derives the Euclidian distance between two positions, $\Delta\mathbb{P}_i = [\Delta P_i^-, \Delta P_i^+]$ and $\Delta P_i^-, \Delta P_i^+$ are the lower and upper bounds of the spacing discrepancy between the position in the equilibrium trajectory and the position in the planned trajectory at time step t .

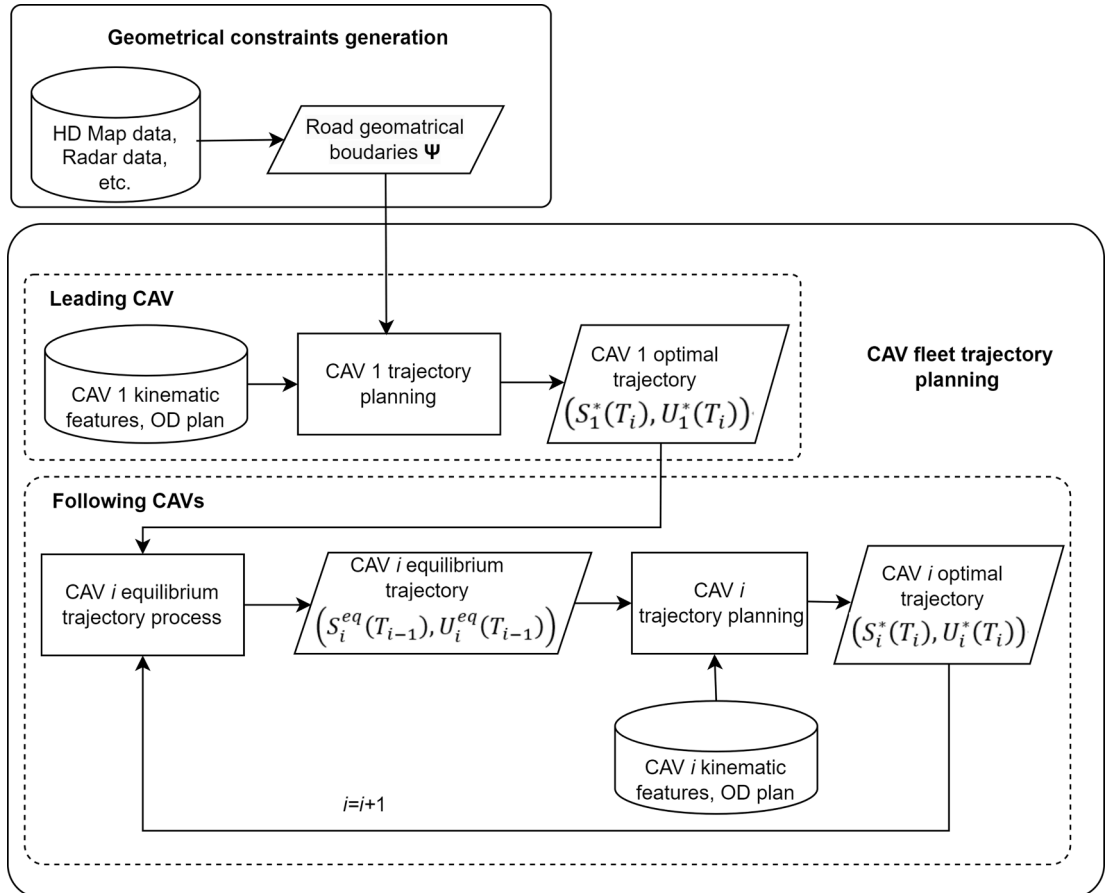


Fig. 4. CAV platoon optimal trajectory planning framework.

Note that the optimization problems in Eq. (5) and Eq. (9) are extremely hard to solve due to Eq. (5-e) and Eq. (9-g). Specifically, $\Psi(z, y)$ can be any available travel area within one/multiple lanes, and it may not be clearly represented by a function with respect to z -axis and y -axis due to irregular boundaries in real world. Even with a clearly represented $\Psi(z, y)$, it presents complexity to the optimal solving algorithms such as the model predictive control (MPC), which is better at handling time-domain constraints rather than spatial-domain constraints such as $\Psi(z, y)$. Furthermore, the nonlinearity of Eq. (5-c), (5-d), (9-e), and (9-f) will significantly increase the solving time and complexity. In this paper, rather than exactly solving the problems as formulated in Eq. (5) and (9), we propose a Cooperative LQR-RRT* algorithm (C-LQR-RRT*) as an approximation. The proposed trajectory planning algorithm is introduced in [section 3](#), where the solving framework and detailed algorithm are elaborated.

3. Solving algorithm

3.1. Solving framework

The solving framework is presented in [Fig. 4](#). It describes how the algorithm will function by collecting road geometric boundaries and preceding CAVs' trajectory portfolios via V2V and V2X communication. With the consideration of road geometry and on-road obstacles (e.g., crashes, construction areas, parking, etc.), the road geometric constraints, represented by Cartesian coordinate, will be processed and transmitted to the first leading CAV in a CAV platoon by a centralized computing unit. Given the geometric constraints, the first leading CAV will generate an optimal trajectory $(S_1^*(T_1), U_1^*(T_1))$. Next, the first leading CAV will transmit its optimal trajectory to its following CAV. After receiving the preceding CAV's optimal trajectory, the following CAV calculates an equilibrium trajectory $(S_i^{eq}(T_{i-1}), U_i^{eq}(T_{i-1}))$ according to the car-following relationship within the CAV platoon. Thereafter, the following CAV will generate an optimal trajectory $(S_i^*(T_i), U_i^*(T_i))$ and transmit to the next following CAV. This process will repeat consecutively until the last CAV in the platoon completes generating its optimal trajectory.

3.2. C-LQR-RRT* algorithm

Based on the framework established above, we propose the Cooperative-LQR-RRT* (C-LQR-RRT*) algorithm to solve the aforementioned problems in [Section 2](#). The C-LQR-RRT* method is an RRT* based method that generates a tree $\mathbb{T}$ rooted from an initial state in $\Psi(z, y) \times \mathbb{M}$ region. The generated tree consists of two sets, a set of feasible vehicle auxiliary states $\mathbb{S}$ and a set of edges $\mathbb{E}$, and $\mathbb{T} = \{\mathbb{S}, \mathbb{E}\}$. States (ξ) in the tree are connected by edges (e), which describe the transitional process from one state to another state. Within the tree, any state can be reached from the initial state through edges via intermediate states. Each edge has a cost according to Eq. (5-a) or (9-a). Therefore, the optimization problem as Eq. (5) and Eq. (9) can be approximated by expanding the tree by adding new states and edges and optimizing the tree topology structure (i.e. the linkage between states), so that a trajectory starting from an initial state to a terminal state with a minimum cost can be found. In addition, Karaman and Frazzoli (2011) proved that the optimal trajectory will converge with an exhausted searching effort.

The C-LQR-RRT* is an extension of the existing method "Poli-RRT*" ([Ragaglia, et al., 2015](#)) and "LQR-RRT*" ([A. Perez et al., 2012](#)) by considering coordination among CAVs. The basic "RRT*" steps are described as follows:

(1) Tree initialization

Two empty sets, $\mathbb{S}$ and $\mathbb{E}$, within tree $\mathbb{T}$ are initialized in this step. The initial auxiliary state of CAV i , $\xi_i(0)$ is added to set $\mathbb{S}$.

(2) Random sampling

The expansion of tree $\mathbb{T}$ starts from randomly sampling a state s_{rand} . In this step, $s_{rand} = [p_{rand}, m_{rand}]^T$ is uniformly sampled within the feasible region $\mathbb{Q}_f \in \Psi(z, y) \times \mathbb{M}$ to guarantee the randomly sampled position is within the feasible road geometry and the motion is within the physical limit. The corresponding auxiliary state ξ_{rand} can be calculated according to Eq. (2).

(3) New state generation

After sampling the random state ξ_{rand} , a nearest state $\xi_{nearest}$ among states $\xi \in \mathbb{S}$ is determined by calculating: $\xi_{nearest} = \underset{\xi \in \mathbb{S}}{\operatorname{argmin}} C(\xi, \xi_{rand})$ through an optimized edge $e(\xi, \xi_{rand})$, where the function $C(\xi, \xi_{rand})$ denotes the cost of traveling along the edge $e(\xi, \xi_{rand})$. A new state ξ_{new} is calculated by moving one time step t_s , the discretization step, forward along the edge $e(\xi_{nearest}, \xi_{rand})$, by which the edge $e(\xi_{nearest}, \xi_{new})$ and the control input at $\xi_{nearest}$, $\eta_{nearest}$ can be derived.

(4) Collision check

An edge $e(\xi, \xi')$ is collision free that $\text{Collisionfree}(e(\xi, \xi')) = \text{True}$ if the edge and all states on the edge is within the region $\Psi(z, y) \times \mathbb{M}$, meanwhile the control inputs at all states on the edge is within the constraint set $\mathbb{U}$. If $\text{Collisionfree}(e(\xi_{nearest}, \xi_{new})) = \text{True}$, $e(\xi_{nearest}, \xi_{new})$ is added to tree $\mathbb{T}$ that $\mathbb{S} = \mathbb{S} \cup \{\xi_{new}\}$ and $\mathbb{E} = \mathbb{E} \cup \{e(\xi_{nearest}, \xi_{new})\}$. Meanwhile, $\xi_{nearest}$ is defined as the parent state of ξ_{new} , that $\text{parent}(\xi_{new}) = \xi_{nearest}$. Otherwise, the generated edges and states are abandoned, and the tree generation process will go back to step 3.

(5) Neighbour state radius computation

This step generates a set of states for limiting the search when optimizing the tree topology. Specifically, this step generates a set of auxiliary states $\mathbb{S}_{near}$ in $\mathbb{S}$ that:

$$\mathbb{S}_{near} := \{\xi | \xi \in \mathbb{S} \wedge |F\xi - F\xi_{new}| < r_{near}\} \quad (10)$$

by a predefined threshold value

$$r_{near} = \gamma^* (\log(N^S) / N^S)^{1/\dim(S)} \quad (11)$$

The threshold r_{near} is designed as a function with respect to the number of existing states within tree $\mathbb{T}(N^S)$, the dimension of states in set S ($\dim(S)$), and a predefined constant γ . Note that the shrinking nature of coefficient $(\log(N^S) / N^S)^{1/\dim(S)}$ is beneficial for an efficient search, whereas a sufficiently large value of constant γ guarantees the convergence of the RRT-based algorithm with an infinite searching effort. The value selection criterion is discussed in detail in Karaman & Frazzoli, (2011).

Fig. 5 demonstrates an example of finding set S_{near} . In Fig. 5, state 1 is ξ_{rand} , state 2 is $\xi_{nearest}$, state 3 is ξ_{new} and the grey circle as a center of state 3 with a radius of r_{near} is the calculated neighbour area by Eq. (10). In this case, $S_{near} = \{state2, state4, state5, state6\}$

(6) Reselecting parent state of the new state:

This step calculates $C(\xi, \xi_{new})$ for $\forall \xi \in S_{near}$ through an optimized edge $e(\xi, \xi_{new})$. For $\xi = \underset{\xi \in S_{near}}{\operatorname{argmin}} C(\xi, \xi_{new})$, if $C(\xi, \xi_{new}) < C(\xi_{nearest}, \xi_{new})$, and $\text{Collisionfree}(e(\xi, \xi_{new})) = \text{True}$, the edge $e(\xi_{nearest}, \xi_{new})$ is removed from $\mathbb{E}$; meanwhile the edge $e(\xi, \xi_{new})$ is added to $\mathbb{E}$, and all states between ξ and ξ_{new} along $e(\xi, \xi_{new})$ is added to set S . Hence, we switch the parent state of ξ_{new} from $\xi_{nearest}$ to ξ , that $\text{parent}(\xi_{new}) = \xi$.

(7) Rewiring tree

This step calculates $C(\xi_{new}, \xi)$ for $\forall \xi \in S_{near}$ through an optimized edge $e(\xi_{new}, \xi)$. For $\xi = \underset{\xi \in S_{near}}{\operatorname{argmin}} C(\xi_{new}, \xi)$, If $C(\xi_{new}, \xi) + C(\xi_{new}, \xi) < C(\xi_{new}, \xi)$ and $\text{Collisionfree}(e(\xi_{new}, \xi)) = \text{True}$, the edge $e(\text{parent}(\xi), \xi)$ is removed from set $\mathbb{E}$; meanwhile the edge $e(\xi_{new}, \xi)$ is added to set $\mathbb{E}$, and all states between ξ_{new} and ξ along $e(\xi_{new}, \xi)$ is added to set S . Note that $C(\xi_{new}, \xi)$ denotes the cost from $\xi_i(0)$ to ξ through existing edges in set $\mathbb{E}$, and correspondingly, we let $e(\xi_{new}, \xi)$ as the cumulative path from $\xi_i(0)$ to ξ . Once finding ξ , the parent state of ξ is updated to ξ_{new} , i.e., $\text{parent}(\xi) = \xi_{new}$. The tree topology optimization process are presented in Fig. 6.

(8) Termination

Step 2–7 is reiteratively implemented until the process reaches the maximum iteration or meets the termination criterion that $\exists \xi \in S, |F_{\xi_{i,end}} - F_{\xi}| < l_{tol}$, where l_{tol} is the tolerance level in distance. The found state ξ is set as $\xi_{i,end}$.

(9) Optimal trajectory generation

The edge $e(\xi_{i,end})$ connecting $\xi_i(0)$ and $\xi_{i,end}$ is returned with the entire tree $\mathbb{T}$, the sequence of states on edge $e(\xi_{i,end})$ is regarded as the optimal trajectory $\{\xi_i(0), \dots, \xi_i^*(T_i)\}$, and a set of optimal control inputs associated with the optimal states, $\{\eta_1^*(0), \dots, \eta_1^*(T_i)\}$ is returned. In addition, all the auxiliary states and control inputs are transformed back to the initially defined states and control inputs by Eq. (2). The set of these transformed states and control inputs are approximations of $S_i^*(T_i)$ and $U_i^*(T_i)$ respectively. The optimal total travel time T_i can be calculated by summing up the travel time through edges $e(\xi_{i,end})$.

The key ingredients of applying the RRT* based algorithm are properly designing the feasible region of sampling ξ_{rand} , $\mathbb{Q}_f$; as well as finding the optimal edge between two states $e(\xi, \xi')$ according to $C(\xi, \xi')$. With respect to the optimization formulations in Eq. (5) and Eq. (9), the aforementioned key ingredients are derived for the leading CAV and the following CAVs respectively in the C-LQR-RRT* algorithm.

C-LQR-RRT* for the leading CAV

According to Eq. (5), the objective of finding optimal trajectory for the leading CAV in a platoon is to minimize the travel time and the control effort. Based on the objective, the key ingredients in C-LQR-RRT* for the leading CAV are defined as follows:

$\mathbb{Q}_f$:

For the leading CAV, the algorithm is designed to explore the entire feasible region and find an optimal trajectory; therefore, the feasible region of generating random states in step (2) is identical to the feasible region of all the states, that $\mathbb{Q}_f = \Psi(z, y) \times \mathbb{M}$

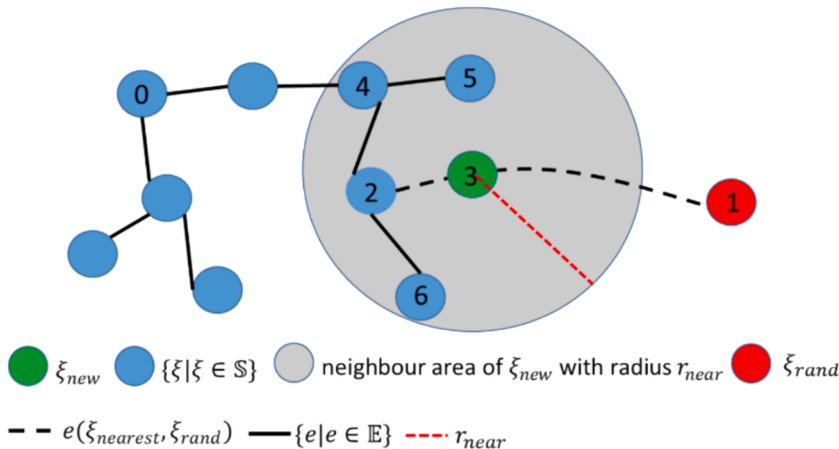


Fig. 5. Finding neighbor state set.

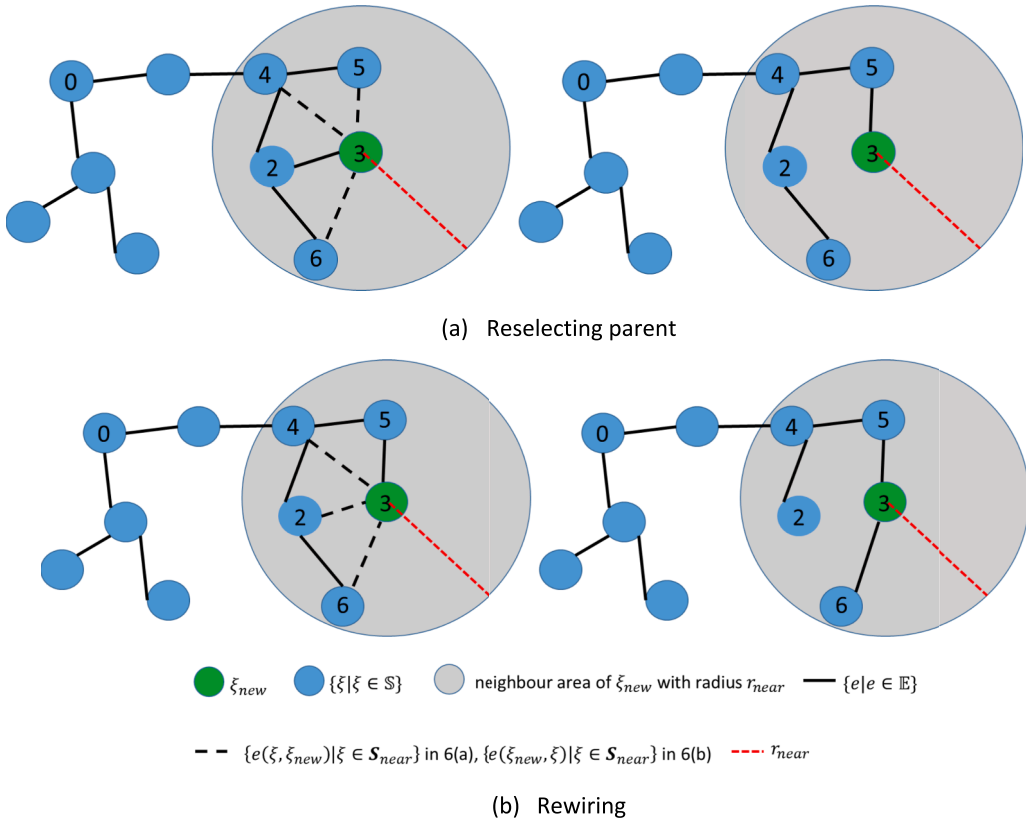


Fig. 6. Tree topology optimization.

Derivation of $e(\xi, \xi')$, $C(\xi, \xi')$ and the travel time:

According to Eq. (5), the cost $C(\xi, \xi')$ can be represented as Eq. (12), given a fixed time τ :

$$C(\xi, \xi', \tau) = \int_0^\tau 1 + \eta(t)^T R \eta(t) dt \quad (12-a)$$

s.t.

$$\frac{d\xi(t)}{dt} = A_r \xi(t) + B_r \eta(t) \quad (12-b)$$

$$\xi(0) = \xi \quad (12-c)$$

$$\xi(\tau) = \xi' \quad (12-d)$$

According to (Webb & Berg, 2012), by fixing τ , the problem of finding the optimal edge $e(\xi, \xi')$ is transferred to a classic linear quadratic “fixed final state and fixed time problem” which can be efficiently solved in a polynomial time by first calculating the weighted controllability Grammian $G(\tau)$ as below:

$$G(\tau) = \int_0^\tau \exp[A_r \times (\tau - t')] B_r R^{-1} B_r^T \exp[A_r^T \times (\tau - t')] dt' \quad (13)$$

Given $G(\tau)$, the optimal control $\eta(t)$ for $\forall t \in [0, \tau]$ can be derived as:

$$\eta(t) = R^{-1} B_r^T \exp[A_r^T \times (\tau - t)] G^{-1}(\tau) \times (-\exp[A_r \times \tau] \times (\xi' - \hat{\xi}(\tau))) \quad (14-a)$$

where $\hat{\xi}(\tau)$ is the state evolution if no control is applied, given as:

$$\hat{\xi}(\tau) = \exp[A_r \times \tau] \xi(0) \quad (14-b)$$

By combining Eq. (12-a) and Eq. (14-a), the optimal cost $C(\xi, \xi', \tau)$ can be written as:

$$C(\xi, \xi', \tau) = \tau + (\xi' - \hat{\xi}(\tau))^T G^{-1}(\tau) (\xi' - \hat{\xi}(\tau)) \quad (15)$$

Then the optimal time τ^* can be calculated by taking the partial derivatives of Eq. (15) on τ , which is given as:

$$\frac{\partial C}{\partial \tau} = 1 - d^T(\tau) B_r R^{-1} B_r^T d(\tau) \quad (16-a)$$

where,

$$d(\tau) = G^{-1}(\tau) (\xi' - \hat{\xi}(\tau)) \quad (16-b)$$

Then we can find the optimal τ^* by

$$\tau^* = \min \tau \quad (17-a)$$

s.t.

$$1 - d^T(\tau^*) B_r R^{-1} B_r^T d(\tau^*) = 0 \quad (17-b)$$

To solve Eq. (17-a), the easiest way is to find all solutions satisfying (17-b) numerically and pick the minimum feasible solution. By which, the optimal control policy can be re-written as Eq. (18) based on Eq. (14):

$$\eta(t) = R^{-1} B_r^T \exp[A_r^T \times (\tau^* - t)] d(\tau^*) \quad \text{for } \forall t \in [0, \tau^*] \quad (18)$$

This results in the optimal state evolution as follows:

$$\xi(t) = [I_4 0_4] \exp \left[\begin{pmatrix} A_r & B_r R^{-1} B_r^T \\ 0_4 & -A_r^T \end{pmatrix} (t - \tau^*) \right] \begin{bmatrix} \xi' \\ d(\tau^*) \end{bmatrix} \quad (19)$$

By Eqs. (12)–(19) optimal edge $e(\xi, \xi')$, the travelling cost $C(\xi, \xi')$ and the travel time τ^* can be derived.

C-LQR-RRT* for the following CAVs

According to Eq. (9), the objective of finding optimal trajectory for the following CAVs in a platoon is to minimize the deviation from their equilibrium trajectory. Given the car-following relationship in Eq. (7) and the optimized arriving time of CAV $i-1$, T_{i-1} , the equilibrium states and control inputs of CAV i , $S_i^{eq}(T_{i-1})$ and $U_i^{eq}(T_{i-1})$, for $i > 1$, can be derived. Based on the equilibrium trajectory, the terminal state of CAV i , $i > 1$, can be derived as $\xi_i^{eq}(T_{i-1})$. The key ingredients in C-LQR-RRT* for the leading CAV are defined as follows:

$\mathbb{Q}_f$:

Based on the optimization objective, the feasible region of generating random states in step (2) for the following CAVs is the states on the equilibrium trajectory, that $\mathbb{Q}_f = S_i^{eq}(T_{i-1}) \times U_i^{eq}(T_{i-1})$, $i > 1$.

$e(\xi, \xi')$, $C(\xi, \xi')$ and the travel time:

For CAV followers, the derivation of $e(\xi, \xi')$ when finding $e(\xi_{nearest}, \xi_{rand})$ in step (3) is different from that of optimizing the tree topology in step (6) and (7).

As the random states is generated on the equilibrium trajectory, which is a sequenced collection of states in time series; therefore, the time step of arriving $\xi_{nearest}$, $t_{nearest}$, should be smaller than the time step of arriving ξ_{rand} , t_{rand} , that $t_{nearest} < t_{rand}$. The problem of finding $e(\xi, \xi_{rand})$, $\xi \in \{\xi | \xi \in \mathbb{S} \wedge t_\xi < t_{rand}\}$ is then transferred to a problem of finding an optimal trajectory in a fixed time horizon $\Delta t = t_{rand} - t_\xi$. Given the new system state as $\bar{\xi}(t) = \xi(t) - \xi^{eq}(t)$ and new control input $\bar{\eta}(t) = \eta(t) - \eta^{eq}(t)$, the cost $C(\xi, \xi_{rand}, \Delta t)$ can be represented by a LQR formulation:

$$C(\xi, \xi_{rand}, \Delta t) = \left(\bar{\xi}(t_{rand})^T Z \bar{\xi}(t_{rand}) + \int_{t_\xi}^{t_{rand}} \bar{\xi}(t)^T Q \bar{\xi}(t) + \bar{\eta}(t)^T R \bar{\eta}(t) dt \right) \quad (20)$$

s.t.

$$\frac{d\bar{\xi}(t)}{dt} = A_r \bar{\xi}(t) + B_r \bar{\eta}(t) \quad (21)$$

where

Q , Z and R^f are defined as in Eq. (9). Note that we set $Z \gg Q$ to ensure $\bar{\xi}(t_{rand})$ approaches zero. Without losing generality, the problem of minimizing $C(\xi, \xi_{rand}, \Delta t)$ can be solved in a polynomial time by a closed form solution, which is in a linear feedback fashion given as:

$$\bar{\eta}(t) = k(t) \bar{\xi}(t) \quad (22)$$

$$k(t) = R_i^{-1} B_r C(t) \quad (23)$$

where $C(t)$ is given by continuous time Recatti differential equation:

$$A_r^T C(t) + C(t)A_r - C(t)B_r R^{-1} B_r^T C(t) + Q = -\dot{C}(t) \quad (24-a)$$

With the boundary condition:

$$C(t_{rand}) = Z \quad (24-b)$$

By which the optimal edge $e(\xi, \xi_{rand})$, the travelling cost $C(\xi, \xi_{rand})$ and the travel time Δt can be derived.

When optimizing the tree topology in step (6) and (7), the optimal edge $e(\xi, \xi')$ is derived by Eq. (12)–(19), while the cost is approximately calculated by Eq. (25) as follows:

$$C(\xi, \xi') = \int_{t_\xi}^{t_{\xi'}} \bar{\xi}(t)^T Q \bar{\xi}(t) + \bar{\eta}(t)^T R \bar{\eta}(t) \quad (25)$$

where $t_{\xi'} = t_\xi + \tau^*$ and τ^* is derived by Eq. (16–17).

4. Experimental results and discussion

4.1. Experiment settings

In this section, a simulation-based experiment is conducted to evaluate the performance of the proposed C-LQR-RRT* algorithm, as the field testing is beyond the scope of this paper. We use a real-world urban road context near the University of Wisconsin-Madison, WI. The road section is 1.4 km long, 10 m wide with horizontal curves, as shown by the yellow line in Fig. 7 (a). There are two lanes and two work zones (marked with construction signs in the figure) in the westbound direction, where the right lane is partially blocked with irregular shaped boundaries. The panoramic view of the simulated context is presented in Fig. 7 (a), and the westbound road geometry based on open street map data (Planet dump retrieved from <https://planet.osm.org>, 2017) is sketched in Fig. 7 (b). In Fig. 7 (b), the solid lines represent feasible road geometrical borders (i.e., road borders and construction area borders; the dashed line is the lane marker, the solid arrow indicates the driving direction, the green circles are starting positions of a CAV platoon; and red circles are the ending positions for the first leading CAV.

In this scenario, the curved road geometry and irregular construction boundaries jointly contribute to the difficulty of trajectory planning for a CAV platoon. The continuity and projection distortion issues of the Frenet coordinate will be amplified. Therefore, it is extremely difficult for existing trajectory planning methods to simultaneously consider geometric and kinematic constraints, as well as the cooperative kinematic dynamics of a CAV platoon.

A platoon of five CAVs is selected for demonstration. Three origin–destination (OD) pairs are simulated: (1) start 1-end 1 with the length of 1.4 km, (2) start 2-end 2 with the length of 0.4 km, and (3) start 3-end 3 with the length of 100 m, as shown in Fig. 6. The CAVs



(a) Experiment scenario panoramic view



(b) Work zones and reshaped lanes

Fig. 7. Experiment scenario.

are expected to follow the car-following relationship defined in section 2 from their starting positions with the initial gap of 20 m. Other than the simulated CAVs, no vehicle is generated in the simulation. We consider six scenarios to examine the proposed algorithm for different lengths of OD pairs, different resolutions (simulation time step length), and different searching efforts. The details of adopted parameters in each scenario are presented in Table 1. Note that the value selection of γ is a trade-off between the computational efficiency and the level of optimization. An overly small selection of this value would result in an infinite searching effort, whereas the optimality cannot be guaranteed with an overly large selection of this value. In this study, the value of γ is related to the planning road length and simulation time step, in order to guarantee a sufficient large size of free space is searched without excessively increasing the computing time.

4.2. Simulation results

Two parts of simulation results are presented in this section. The first part presents the optimal state and control input evolution for all the CAVs in a platoon, and the second part presents the results of traffic efficiency, CAV platoon performance, and computational performance. The simulation was run 10 times with different random seeds. Here we present detailed results from one selected simulation run as an example, except for the computational performance, for which we report the average value from the 10 runs.

State and control input revolution results:

Fig. 8 shows three selected position results for the first leading CAV with different length of planning distances over a 1 s time step. In each sub-figure, the upper part shows the optimal position of the first leading CAV in every simulation step; and the bottom part shows the corresponding lane occupancy along the z-coordinate. The driving direction is indicated by the solid arrow and the work zone is marked. It is evident that the C-LQR-RRT* algorithm is capable of navigating individual CAVs to drive through the complex road geometry with different planning distances. The optimal trajectory follows the terrain with a mild change in road geometry (e.g., see the positions for the first 300 m in Fig. 8-a); whereas it largely follows the tangent of the road boundary with a more drastic change in geometry (e.g., positions around 1000–1200 m in Fig. 8-a). Since the C-LQRRRT* algorithm anticipates the road geometry change, work zones, and the destination while trying to minimize the control effort, the situation of impeding or being impeded by other vehicles while changing lanes is not considered.

Based on the optimal trajectory of the first leading CAV, the optimal trajectories of other CAVs in the same platoon are generated following the framework in section 3. As an illustrative example, Fig. 9-a presents snapshots of the optimized positions of the CAV platoon over 9 consecutive time steps near a work zone in scenario 4. It can be observed that all CAVs were navigated to drive from the right lane to the left lane smoothly with the obstacle ahead. In Fig. 9-b, the lane occupancy for each CAV is presented along the relative z-axis (the top (bottom) plot corresponds to the first (last) CAV). The blue and red diamonds represent the starting and ending positions, respectively. It can be observed that all CAVs changed lanes at approximately the same location around 160 m and 280 m to avoid the work zone, which is suggested by Eq. (6) and Fig. 3.

Fig. 10 presents the motion state and control input evolution in scenario 4. For the first CAV (blue curves), some decelerations and lower speeds are notable when avoiding work zones and changing lanes (at 4–11 s) in Fig. 10-a. Further, other than avoiding work

Table 1
Parameter settings in each examined scenario.

Description	Symbol	Scenarios					
		1	2	3	4	5	6
OD pair		(1)		(2)		(3)	(2)
road length (km)	l_{road}	1.4		0.4		0.1	0.4
simulation time step (s)	t_s	3	1	2	1	1	1
CAV length (m)				5			
CAV number in one platoon	N_{CAV}			5			
CAV speed limit (m/s)	(v_{min}, v_{max})			(8,16)			
CAV direction limit	$(\theta_{min}, \theta_{max})$			$(-\pi, \pi)$			
CAV acceleration limit (m/s ²)	$(\alpha_{min}, \alpha_{max})$			(-2,2)			
CAV direction variation limit	$(\omega_{min}, \omega_{max})$			$(-\pi/6, \pi/6)$			
CAV initial speed (m/s)	$v_i(0)$			11			
CAV initial direction	$\theta_i(0)$			π			
CAV terminal speed (m/s)	$v_i(T_i)$			13			
CAV terminal direction	$\theta_i(T_i)$			π			
terminal state sample rate	σ_{end}			0.3			
minimum gap (m)	c_i			2			
desired time headway (s)	φ_i			0.6			
car-following security threshold	$\Delta P_i = [\Delta P_i^-, \Delta P_i^+]$			$[-\infty, 2]$			
element value in matrix Q	$(\beta_1, \beta_2, \beta_3, \beta_4)$			(0.1,0.1,0.5,0.5)			
element value in matrix Z	(z_1, z_2, z_3, z_4)			(4,4,1,2)			
element value in matrix R	(γ_z, γ_y)			(1,2)			
element value in matrix R ^f	(γ_z^f, γ_y^f)			(1,2)			
constant of neighbor states radius in step 5	γ	120	420	60	120	30	120
max iteration				-			3000
if algorithm run until max iteration				No			Yes

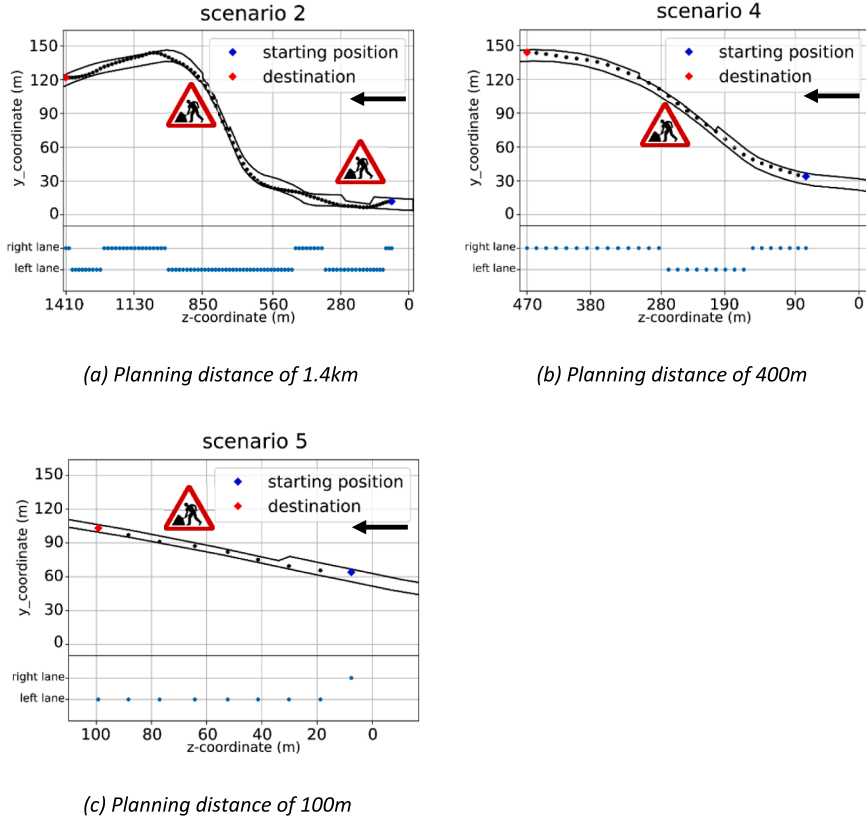


Fig. 8. Trajectory and lane occupancy of CAV 1 for different planning distances over a 1 s time step.

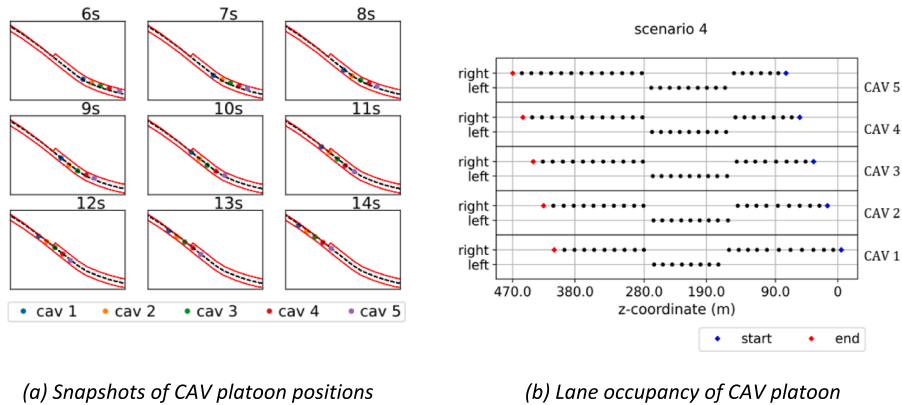
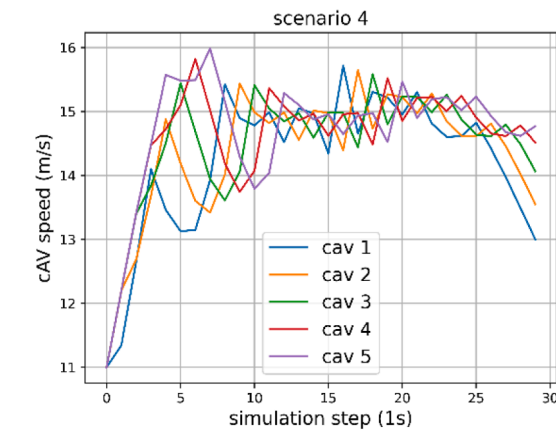


Fig. 9. CAV platoon positions and lane occupancy in scenario 4.

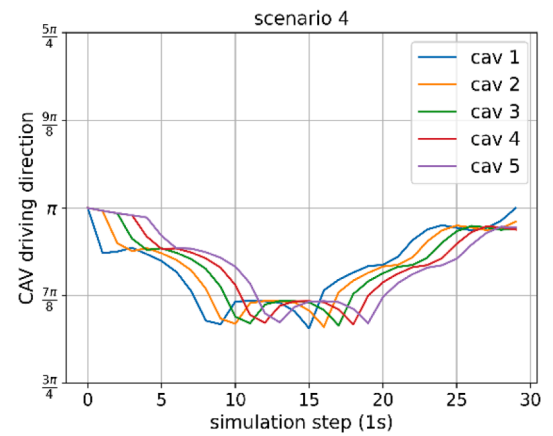
zones, the first CAV was controlled to alter its driving direction following the road geometry; see Fig. 10-b. The following CAVs initially accelerate to close the gaps and engage in a car-following relationship in approximately the first 10 s. Thereafter, their speeds and driving directions are synchronized with their predecessors with some latency. The control effort evolution in scenario 4 are presented in Fig. 10-c and Fig. 10-d. It can be observed that the oscillations in longitudinal and angular accelerations can be dampened through CAVs in a platoon in all scenarios, which suggests the empirical string stability of our proposed method.

Evaluation results

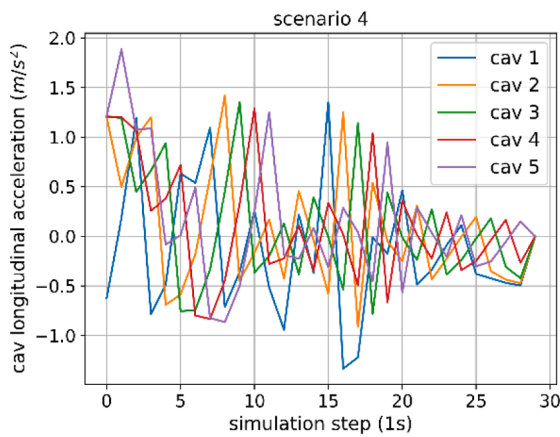
To systematically evaluate the performance of the proposed algorithm, we use traffic efficiency, string stability, and computational efficiency as criteria. The traffic efficiency is evaluated by the normalized travel time delay:



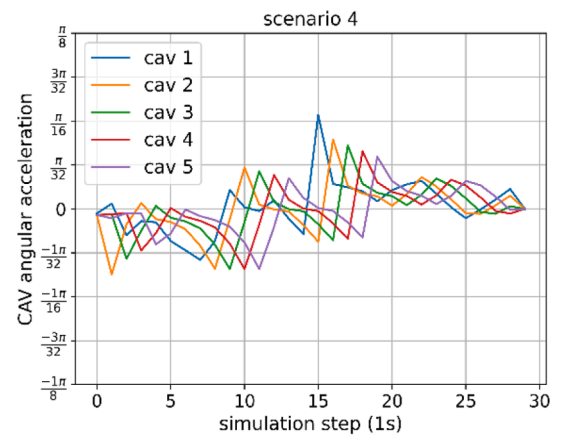
(a) speed



(b) driving direction



(c) longitudinal acceleration



(d) angular acceleration

Fig. 10. Motion state and control input evolution for cavs in scenario 4.

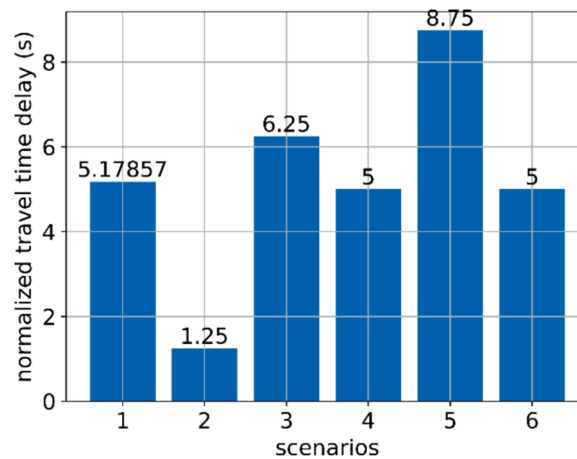


Fig. 11. Travel time delay results.

$$T^{\text{delay}} = \sum_{i=1}^{i=N_{\text{CAV}}} \left(T_i - \frac{l_{\text{road}}}{v_{\text{max}}} \right) / \left(\frac{l_{\text{road}}}{100} \right) \quad (33)$$

The travel time delays for all six scenarios are presented in Fig. 11. In general, the travel time delay decreases with a higher resolution (scenario 1 vs. scenario 2, scenario 3 vs. scenarios 4), whereas applying a maximum iteration in the algorithm has a minor impact (scenario 4 vs. scenario 6). In addition, applying a short planning distance (e.g., 100 m in scenario 5) is less efficient for reducing travel time delay.

The second evaluation criterion, string stability, is an important property for an interconnected system to guarantee disturbance attenuation. In our framework, the longitudinal acceleration $a_i(t)$ and the angular acceleration $\omega_i(t)$ are considered to represent external disturbances. Considering the peak magnitude and the energy of disturbances, we extend the l_∞ and l_2 string stability definitions by Naus et al. (Naus et al., 2010) into two dimensions as below:

Definition 1: Two dimensional l_∞ String Stability: For a system of a CAV platoon with size N_{CAV} , it is l_∞ string stable if and only if:

$$|\alpha_{i+1}(t)|_{l_\infty} \leq |\alpha_i(t)|_{l_\infty} \text{ for } \forall i \in [1, 2, \dots, N_{\text{CAV}} - 1]$$

and

$$|\omega_{i+1}(t)|_{l_\infty} \leq |\omega_i(t)|_{l_\infty} \text{ for } \forall i \in [1, 2, \dots, N_{\text{CAV}} - 1]$$

Definition 2: Two dimensional l_2 String Stability: For a system of a CAV platoon with size N_{CAV} , it is l_2 string stable if and only if:

$$|\alpha_{i+1}(t)|_{l_2} \leq |\alpha_i(t)|_{l_2}, \text{ for } \forall i \in [1, 2, \dots, N_{\text{CAV}} - 1]$$

and

$$|\omega_{i+1}(t)|_{l_2} \leq |\omega_i(t)|_{l_2}, \text{ for } \forall i \in [1, 2, \dots, N_{\text{CAV}} - 1]$$

The result of the l_∞ and l_2 string stability analyses is presented in Fig. 12. Specifically, the damping rate (i.e., the ratio of following CAV's peak magnitude and energy to the first leading CAV) in all six examined scenarios are presented. Note that as mentioned in the discussion of Fig. 10, the following CAVs initially accelerate to catch up and engage in car-following. Therefore, the data before the following CAV first reaching the desired headway (i.e., φ_i) is omitted. In addition, the string stability performance of a recent state-of-the-art algorithm (Zhang et al., 2020) based on the Frenet coordinate modeling is examined in scenario 4 for comparison. The results show that both the peak magnitude and the energy of disturbances (l -infinity norm and l -2 norm, respectively) diminish through the CAV string in all six scenarios with the proposed algorithm (damping rates < 1). In contrast, disturbances grow with the algorithm based on the Frenet coordinate (damping rates > 1). Clearly, the C-LQR-RRT* algorithm is capable of planning the optimal trajectory for the CAV platoon while achieving empirical string stability, which cannot be ensured by the state-of-the-art Frenet coordinate based algorithm.

Finally, the computational inference time is presented in Table 2. The simulations were coded in Python 3.7 and operated by a laptop with Intel Core i5-6300HQ 2.30 Hz CPU and 8 GB memory. The results show that scenario 5 has the fastest average computational time of 2.4 s for all CAVs. As expected, the average computational time increases with increased OD pair distance (e.g., scenario 2 vs. scenario 4), higher resolution (e.g., scenario 3 vs. scenario 4), and greater searching effort (e.g., scenario 4 vs. scenario 6). The results indicate that the computational time is primarily determined by the number of iterations, rather than having a linear relationship with the OD pair distance or the resolution. Consequently, increasing the number of CAVs in a platoon would not significantly impact computational efficiency. Overall, the proposed algorithm is capable of planning trajectories of a CAV platoon within reasonable time frames, given reasonable planning distance (< 1 km) and control resolution (> 1 s). This adaptability suggests two potential applications: implementation with a rolling-horizon framework to generate smoother trajectories, and handling of unexpected situations (e.g., unexpected accidents) by planning over shorter distances.

5. Conclusion and discussion

This paper presented a cooperative two-dimensional trajectory planning algorithm for CAV platoons. Specifically, the proposed algorithm generated two-dimensional optimal trajectories for CAVs cooperatively under a curved road geometry and with complex urban settings. By extending the simplified Newell's car-following model, we proposed a two-dimensional Newell's car-following model as an equilibrium car-following policy for CAVs. Based on that, multi-objective constrained optimizations were formulated in a Cartesian coordinate. Due to the constraint complexity, a new sampling-based solving algorithm based on the rapid random tree technique was proposed to solve the problem efficiently. The proposed algorithm is a product of an interdisciplinary effort, integrating the widely used RRT-based algorithm in robotics, the optimal control theory, and the car-following model in traffic flow theory. Through simulations, we examined that the proposed algorithm is capable of generating optimal trajectories for CAV platoons with a short travel time delay and optimized control efforts. Notably, this algorithm is applicable with any complicated road geometric constraints, incorporating irregular shaped boundaries of road-side parkings, temporary traffic regulations, or incidents, which was lacking in traditional methods.

In addition, the simulation results showed that with a sophisticated design of control methods and trajectory searching strategies, the proposed algorithm achieved a promising result with moderate computational effort. These results indicate that the C-LQR-RRT* algorithm has the potential to be implemented in practice.

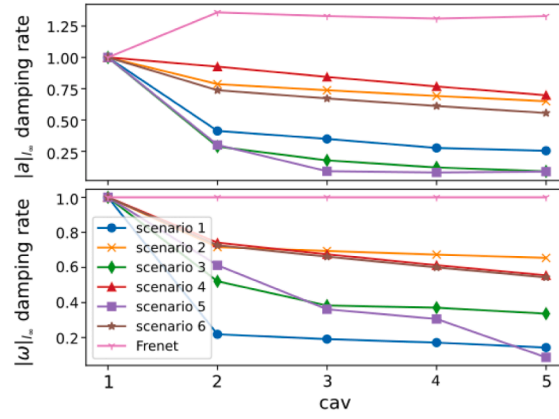
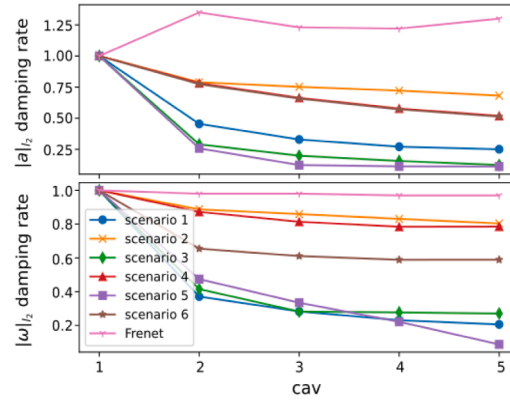
(a) l_∞ norm damping rate(b) l_2 norm damping rate

Fig. 12. String stability test results for CAV platoon.

Table 2

Average computational time for different scenarios.

Scenarios	1	2	3	4	5	6
CAV leader compute time (s)	10.2	52.6	4.7	8.9	1.3	589
CAV followers compute time (s)	3.9	5.8	1.3	3.7	1.1	2401

However, the algorithm can be less effective with long-distance OD pairs and small discretization steps. These deficiencies can be solved through V2V communication and a rolling horizon planning framework, which can be investigated in depth in the future. In addition, while the proposed algorithm is dedicated to deal with curved roads with static and irregular shaped obstacles in complex urban settings, handling dynamic obstacles will require adopting a hierarchical and iterative framework, making a major extension in the future. It is well-established that centralized control can yield better control performance than distributed control. Thus, we suggest that future studies expand the algorithm in a centralized fashion to further explore the benefits of the proposed framework. We caution, however, that distributed control may be more realistic for traffic systems as it is difficult to implement centralized decision-making. Finally, as the computational time heavily relies on launched iterations in the algorithm, the computational efficiency may be improved by reducing the searching randomness. This could potentially be achieved by coupling feasible motion states with position states through reinforcement learning.

CRedit authorship contribution statement

Shengyue Yao: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. Yang

Zhou: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization. **Bernhard Friedrich:** Writing – review & editing, Supervision, Conceptualization. **Soyoung Ahn:** Writing – review & editing, Supervision, Conceptualization.

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